

# Cap-and-Trade and Public Health: Evidence from New Jersey's Exit and Reentry into the Regional Greenhouse Gas Initiative

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May 2026

# Introduction

## Greenhouse Gases (GHGs)

- Gases that maintain Earth's stable atmospheric temperatures (Kweku et al., 2018)
- Carbon dioxide ( $CO_2$ ), nitrous oxide ( $N_2O$ ), and methane ( $CH_4$ ) (Filonchyk et al., 2024)

## Regional Greenhouse Gas Initiative

- Regulates carbon dioxide emissions from electrical power plants (RGGI, 2026)
- Maine, New Hampshire, Vermont, Connecticut, New York, Delaware, and New Jersey, Massachusetts, Maryland, and Rhode Island (2009)
- Revenue generation, emissions reductions, and leakages

# Data: Emissions

## Source

- EPA's Inventory of U.S. Greenhouse Gas Emissions and Sinks by State: 1990-2022 (EPA, 2022)
- "requires reporting of greenhouse gas (GHG) data and other relevant information from large GHG emission sources, fuel and industrial gas suppliers, and CO2 injection sites in the United States" (EPA, 2025)

## Units and Important Points

- Total emissions are measured in Teragrams (trillions of grams)
- 2009 (RGGI launch), 2012 (NJ exit), 2020 (NJ reentry)

# Data: Emissions Summary

**Table 2: Power Sector CO<sub>2</sub> Emissions by RGGI State<sup>1</sup>**

*Total emissions in teragrams (Tg CO<sub>2</sub>), 2000–2022*

| State     | Pre-RGGI     |              |              | RGGI Launch  |              | NJ Exit Period |              | NJ Reentry   |              |
|-----------|--------------|--------------|--------------|--------------|--------------|----------------|--------------|--------------|--------------|
|           | 2000         | 2003         | 2006         | 2009         | 2012         | 2015           | 2018         | 2020         | 2022         |
| CT        | 11.82        | 8.79         | 10.81        | 7.74         | 8.42         | 8.60           | 9.00         | 9.73         | 10.16        |
| DE        | 5.39         | 5.99         | 5.85         | 3.94         | 4.65         | 3.35           | 2.59         | 1.88         | 2.08         |
| MA        | 23.82        | 26.52        | 23.54        | 19.39        | 13.68        | 13.04          | 9.22         | 7.24         | 8.35         |
| MD        | 32.32        | 33.12        | 30.76        | 25.68        | 20.24        | 17.86          | 17.27        | 10.06        | 11.31        |
| ME        | 3.97         | 5.10         | 3.18         | 2.81         | 2.18         | 1.98           | 1.37         | 1.00         | 2.00         |
| NH        | 4.86         | 7.50         | 7.21         | 5.69         | 4.47         | 3.84           | 2.31         | 1.82         | 2.66         |
| <b>NJ</b> | <b>20.03</b> | <b>18.92</b> | <b>19.54</b> | <b>15.87</b> | <b>16.00</b> | <b>19.08</b>   | <b>18.50</b> | <b>14.62</b> | <b>15.44</b> |
| NY        | 58.83        | 55.42        | 49.62        | 36.77        | 34.42        | 31.33          | 26.50        | 25.65        | 29.27        |
| RI        | 2.68         | 2.32         | 2.36         | 3.05         | 3.40         | 2.87           | 3.20         | 3.36         | 2.83         |
| VT        | 0.26         | 0.14         | 0.15         | 0.14         | 0.23         | 0.26           | 0.22         | 0.23         | 0.25         |

<sup>1</sup> Note: New Jersey (highlighted) exited RGGI in 2012 and rejoined in 2020. Data source: EPA State GHG Inventory 1990–2022.

# Data: Health

## Source

- Behavioral Risk Factor Surveillance System (BRFSS) Prevalence Data (CDC)
- Questionnaire is a “random-digit-dialed telephone survey of the non-institutionalized civilian population 18 years of age and older” (CDC, 2023)

## Units and Important Points

- Asthma prevalence is measured in percent of adults with asthma in sample
- 2009 (RGGI launch), 2012 (NJ exit), 2020 (NJ reentry)

# Data: Health Summary

**Table 3: Adult Asthma Prevalence by RGGI State<sup>1</sup>**

*Current asthma prevalence (%), 2000–2022*

| State     | Pre-RGGI     |              |              | RGGI Launch  |              | NJ Exit Period |              | NJ Reentry   |              |
|-----------|--------------|--------------|--------------|--------------|--------------|----------------|--------------|--------------|--------------|
|           | 2000         | 2003         | 2006         | 2009         | 2012         | 2015           | 2018         | 2020         | 2022         |
| CT        | 8.34%        | 8.56%        | 9.64%        | 9.06%        | 9.71%        | 10.34%         | 10.34%       | 10.51%       | 11.97%       |
| DE        | 7.40%        | 8.01%        | 9.47%        | 8.55%        | 9.73%        | 9.45%          | 9.68%        | 9.96%        | 10.87%       |
| MA        | 9.07%        | 10.12%       | 11.07%       | 11.37%       | 11.17%       | 10.68%         | 10.41%       | 10.87%       | 11.45%       |
| MD        | 7.56%        | 7.69%        | 8.56%        | 8.83%        | 8.54%        | 9.00%          | 9.24%        | 9.31%        | 10.56%       |
| ME        | 8.99%        | 9.89%        | 10.20%       | 10.56%       | 10.80%       | 10.85%         | 11.32%       | 10.73%       | 12.11%       |
| NH        | 8.80%        | 8.82%        | 10.15%       | 10.21%       | 9.80%        | 10.05%         | 10.79%       | 10.96%       | 11.61%       |
| <b>NJ</b> | <b>6.70%</b> | <b>7.33%</b> | <b>7.84%</b> | <b>8.08%</b> | <b>8.95%</b> | <b>7.74%</b>   | <b>9.44%</b> | <b>9.09%</b> | <b>8.66%</b> |
| NY        | 8.26%        | 7.96%        | 8.82%        | 9.92%        | 9.46%        | 9.92%          | 10.84%       | 9.68%        | 10.09%       |
| RI        | 9.02%        | 10.35%       | 10.68%       | 11.11%       | 11.14%       | 11.25%         | 12.25%       | 12.88%       | 12.66%       |
| VT        | 7.39%        | 8.08%        | 9.32%        | 9.81%        | 10.00%       | 10.52%         | 11.36%       | 10.00%       | 11.99%       |

<sup>1</sup> Note: New Jersey (highlighted) exited RGGI in 2012 and rejoined in 2020. Data source: CDC Behavioral Risk Factor Surveillance System (BRFSS), 2000–2022.

# Methods: Emissions

## Differences-in-Differences

$$\begin{aligned} \text{Emissions}_{st} = & \alpha + \beta_1(\text{NJExit}_s \times \text{Post}_{2012,t}) \\ & + \beta_2(\text{NJReentry}_s \times \text{Post}_{2020,t}) + \gamma_s + \delta_t + \varepsilon_{st} \end{aligned} \quad (1)$$

## Variable Descriptions

- $\text{Emissions}_{st}$  — power sector CO<sub>2</sub> emissions for state  $s$  in year  $t$
- $\beta_1$  — effect of NJ exit on emissions relative to core RGGI states post-2012
- $\beta_2$  — effect of NJ reentry on emissions relative to core RGGI states post-2020
- $\gamma_s$  — state fixed effects
- $\delta_t$  — year fixed effects
- $\varepsilon_{st}$  — error term

# Methods: Health

## Differences-in-Differences

$$\text{Asthma}_{st} = \alpha + \beta_1(\text{NJExit}_s \times \text{Post}_{2012,t}) + \beta_2(\text{NJReentry}_s \times \text{Post}_{2020,t}) + \gamma_s + \delta_t + \varepsilon_{st} \quad (2)$$

## Variable Descriptions

- $\text{Asthma}_{st}$  — current adult asthma prevalence for state  $s$  in year  $t$
- $\beta_1$  — effect of NJ exit on asthma prevalence relative to core RGGI states post-2012
- $\beta_2$  — effect of NJ reentry on asthma prevalence relative to core RGGI states post-2020
- $\gamma_s$  — state fixed effects
- $\delta_t$  — year fixed effects
- $\varepsilon_{st}$  — error term



# Findings

## Emissions

- 2012: New Jersey's CO<sub>2</sub> emissions increased by 4.54% in comparison to the core RGGI states (5% significance)
- 2020: New Jersey's CO<sub>2</sub> emissions declined by 0.916% in comparison to that of the RGGI core states (not significant)

## Health

- 2012: New Jersey's asthma rate increased by 0.244% relative to core RGGI states (not significant)
- 2020: New Jersey's asthma rate decreased by 0.524% relative to the core RGGI states (5% significance)

# Findings Table

**Table 1: Two-Sided DiD Results<sup>1,2</sup>**

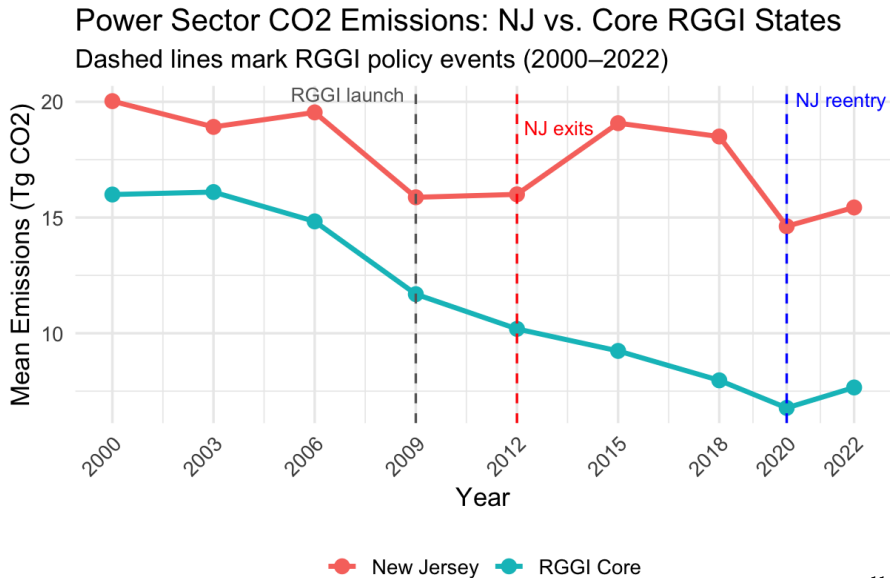
*NJ RGGI Exit (2012) and Reentry (2020)*

|                          | CO <sub>2</sub> Emissions (Tg) | Asthma Prevalence (%) |
|--------------------------|--------------------------------|-----------------------|
| <b>NJ Exit (2012)</b>    | <b>4.544**</b>                 | <b>0.244</b>          |
|                          | (1.190)                        | (0.195)               |
| <b>NJ Reentry (2020)</b> | <b>-0.916</b>                  | <b>-0.524*</b>        |
|                          | (0.979)                        | (0.164)               |
| State FE                 | Yes                            | Yes                   |
| Year FE                  | Yes                            | Yes                   |
| Observations             | 60                             | 60                    |
| Within R <sup>2</sup>    | 0.053                          | 0.028                 |
| Symmetry Test p-value    | —                              | 0.323                 |

<sup>1</sup> Note: Standard errors in parentheses, clustered at the state level. Control group: CT, DE, ME, MD, MA, NH, NY, RI, VT. Sample restricted to 2009 onwards. Symmetry test:  $H_0: \beta_1 = -\beta_2$ .

<sup>2</sup> \* p<0.05 \*\* p<0.01 \*\*\* p<0.001

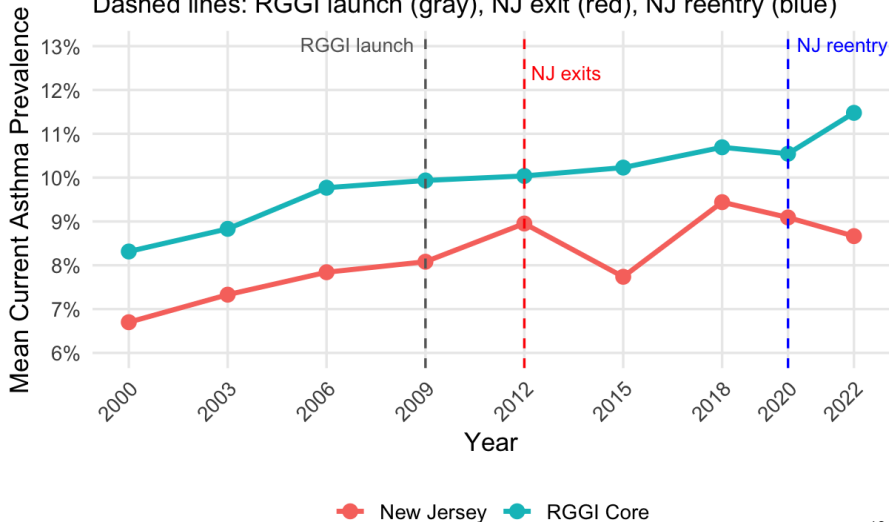
# Findings Graph: Emissions



# Findings Graph: Health

## Adult Asthma Prevalence: NJ vs. Core RGGI States

Dashed lines: RGGI launch (gray), NJ exit (red), NJ reentry (blue)



# What Did We Learn?

- Cap and trade programs can produce more than just revenue
- Deregulation has a positive relationship with increased CO<sub>2</sub> emissions
- Regulation has a negative relationship with asthma prevalence rates among adults

# References I

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# References II

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